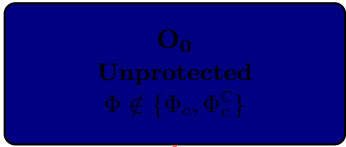
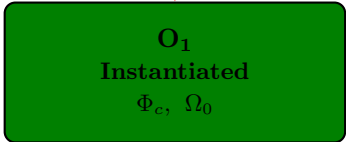


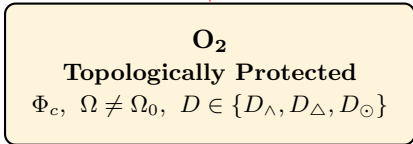
Tier structure



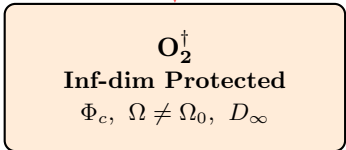
$$\Delta = 1.05 \Phi_{\text{sub}} \rightarrow \Phi_c$$



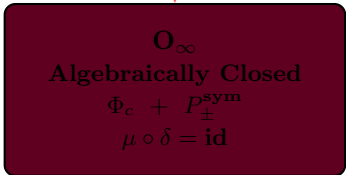
$$\Delta = 1.30 D_{\wedge} \rightarrow D_{\Delta}, \Omega_0 \rightarrow \Omega_{\mathbb{Z}_2}$$



$$\Delta = 1.00 D_{\Delta} \rightarrow D_{\infty}$$



$$\Delta = 4.38 P_{\text{asym}} \rightarrow P_{\pm}^{\text{sym}} \text{ Largest single-primitive jump}$$



4 gate primitives:
L4 fails Φ (L4: Lawvere)
No self-modeling loop P (L5: Frobenius)
C = 0 (Gate 1 fails) D (I4: Yoneda)
 Ω (I3: Classifying)

8 shape primitives:
 $T, R, F, K, G, \Gamma, H, S$

Tier boundaries: *exclusively*
L4 holds $\Phi + P$ promotions
L5 fails
Loop forms, no winding protection

Winding-number protection
Finite-dim or imscriptive
 $C > 0$ possible

Infinite-dimensional
Winding protection
Not imscriptive

Both L4 and L5 hold
Frobenius exact
Self-encoding fixed point
C: Gate 2 open